

Chapter Seven

Method of Undetermined Coefficients

Section 7.1

Method of Undetermined Coefficients (Superposition Approach)

7.1.1 Method of Undetermined Coefficients

Let us consider the non-homogeneous differential equation

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = F(x) \quad \dots\dots(1)$$

where the coefficients $a_0, a_1, \dots, a_n$ are constants but where the non-homogeneous term F is (in general) a non-constant function of x . The general solution of (1) may be written as

$$y = y_c + y_p \quad \dots\dots(2)$$

where y_c is the complementary function, that is, the general solution of the corresponding homogeneous equation (1) and y_p is a particular integral, that is, any solution of (1) containing no arbitrary constants.

To find particular integral, y_p , we consider first the method of undetermined coefficients.

Definition: We shall call a undetermined coefficients (UC) function if it is either (A) a function defined by one of the following:

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| (i) x^n , where n is a positive integer or zero,
(ii) e^{ax} , where a is a constant not equal zero ($\neq 0$), | (iii) $\sin(bx+c)$, where b and c are constants, $b \neq 0$,
(iv) $\cos(bx+c)$, where b and c are constants, $b \neq 0$. |
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or (B) a function defined as a finite product of two or more functions of these four types.

Examples:

- (A) UC functions x^4 , e^{2x} , $\sin(2x+3)$, $\cos(5x/2)$ are respectively the four basic types stated in (i), (ii), (iii) and (iv) of the previous definition.
- (B) UC functions of finite products of two or more of these four basic types are those of defined respectively by $x^3 e^{2x}$, $x \sin 2x$, $e^{3x} \cos 3x$, $\sin 2x \cos 4x$, $x^2 e^{3x} \sin 4x$.

The method of undetermined coefficients applies when the non-homogeneous function F in the differential equation is a finite linear combination of UC functions.

Undetermined coefficients (UC) Set of function: Consider a UC function f . The set of functions consisting of f itself and all linearly independent UC functions of which the successive derivatives of f are either constant multiples or linear combinations will be called the UC set of f .

Examples:

- (i) Let us consider the UC function $f = x^3$ ($\forall x \in \mathbb{R}$). Computing the derivative of $f(x)$, we find $f'(x) = 3x^2$, $f''(x) = 6x$, $f'''(x) = 6 \cdot 1$, $f^{(n)}(x) = 0$ for $n > 0$.

The linearly independent UC functions of which the successive derivatives of f are either constant multiples or linear combinations are those given by x^2 , x , 1 .

Thus the UC set of x^3 is the set $S = \{x^3, x^2, x, 1\}$.

- (ii) Let us consider the UC function $f = \cos 2x$ ($\forall x \in \mathbb{R}$). Computing the derivative of $f(x)$, we find $f'(x) = -2\sin 2x$, $f''(x) = -2^2 \cos 2x$, $f'''(x) = 2^3 \sin 2x$, ...

The only linearly independent UC function of which the successive derivatives of f are either constant multiples or linear combinations is that given by $\sin 2x$, $\cos 2x$.

Thus the UC set of $\cos 2x$ is the set $S = \{\sin 2x, \cos 2x\}$.

- (iii) Let us consider the UC function $f = x^2 e^{2x}$ ($\forall x \in \mathbb{R}$). Computing the derivative of $f(x)$, we find

$$f'(x) = 2x^2 e^{2x} + 2xe^{2x},$$

$$f''(x) = 4x^2 e^{2x} + 8xe^{2x} + 2e^{2x},$$

$$f'''(x) = 8x^2 e^{2x} + 24xe^{2x} + 12e^{2x},$$

No "new" types of functions will occur from further differentiation. Each derivative of f is a linear combination of certain of the three UC functions given by $x^2 e^{2x}$, xe^{2x} , e^{2x} .

Thus the UC set of $x^2 e^{2x}$ is the set $S = \{x^2 e^{2x}, xe^{2x}, e^{2x}\}$.

The UC set for some UC functions are given in the following table:

Sl No.	UC function	UC Set
1.	x^n	$\{x^n, x^{n-1}, x^{n-2}, \dots, x, 1\}$
2.	e^{ax}	$\{e^{ax}\}$
3.	$\sin(bx+c)$ or $\cos(bx+c)$	$\{\sin(bx+c), \cos(bx+c)\}$
4.	$x^n e^{ax}$	$\{x^n e^{ax}, x^{n-1} e^{ax}, x^{n-2} e^{ax}, \dots, x e^{ax}, e^{ax}\}$
5.	$x^n \sin(bx+c)$ or $x^n \cos(bx+c)$	$\{x^n \sin(bx+c), x^n \cos(bx+c), x^{n-1} \sin(bx+c), x^{n-1} \cos(bx+c), x^{n-2} \sin(bx+c), x^{n-2} \cos(bx+c), \dots, x \sin(bx+c), x \cos(bx+c), \sin(bx+c), \cos(bx+c)\}$
6.	$e^{ax} \sin(bx+c)$ or $e^{ax} \cos(bx+c)$	$\{e^{ax} \sin(bx+c), e^{ax} \cos(bx+c)\}$
7.	$x^n e^{ax} \sin(bx+c)$ or $x^n e^{ax} \cos(bx+c)$	$\{x^n e^{ax} \sin(bx+c), x^n e^{ax} \cos(bx+c), x^{n-1} e^{ax} \sin(bx+c), x^{n-1} e^{ax} \cos(bx+c), x^{n-2} e^{ax} \sin(bx+c), x^{n-2} e^{ax} \cos(bx+c), \dots, x e^{ax} \sin(bx+c), x e^{ax} \cos(bx+c), e^{ax} \sin(bx+c), e^{ax} \cos(bx+c)\}$

Example-1 Solve the differential equation by the method of undetermined coefficients:

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x$$

Solution Given differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x$ (1)

The corresponding homogeneous equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 0$ (2)

Let $y = e^{mx}$ be a trial solution of the equation (2), then the auxiliary equation is

$$\begin{aligned} m^2 - 2m - 3 &= 0 \\ \Rightarrow (m-3)(m+1) &= 0 \\ \therefore m &= 3, -1 \end{aligned}$$

Thus the complementary function of (1) is $y_c = c_1e^{3x} + c_2e^{-x}$ (3)

The non-homogeneous term $F(x) = 2e^x - 10\sin x$ is the linear combination of the two UC functions given by e^x and $\sin x$.

- Form the UC set for each of these two functions. We find $S_1 = \{e^x\}$, $S_2 = \{\sin x, \cos x\}$
- From the above UC sets we observe that no element is common of these two functions. Hence both functions are retained.
- Furthermore, by examining the complementary function, we see that none of the functions e^x , $\sin x$, $\cos x$ in either of these sets is a solution of the corresponding homogeneous equation. Hence neither set needs to be revised.

Thus the final UC set is the union of S_1 and S_2 , that is, $S = \{e^x, \sin x, \cos x\}$.

Let the particular solution of (1) be

$$y_p = Ae^x + B\sin x + C\cos x \quad \text{.....(4)}$$

Then,

$$\begin{aligned} y'_p &= Ae^x + B\cos x - C\sin x \\ y''_p &= Ae^x - B\sin x - C\cos x \end{aligned}$$

Substituting these in (1) we obtain,

$$\begin{aligned} (Ae^x - B\sin x - C\cos x) - 2(Ae^x + B\cos x - C\sin x) - 3(Ae^x + B\sin x + C\cos x) &= 2e^x - 10\sin x \\ \Rightarrow -4Ae^x + (-4B + 2C)\sin x + (-4C - 2B)\cos x &= 2e^x - 10\sin x \end{aligned}$$

The above relation is an identity for all x and hence equating the coefficients of like terms on both sides, we get $-4A = 2$, $-4B + 2C = -10$, $-4C - 2B = 0$

Solving above equations, we find that $A = -\frac{1}{2}$, $B = 2$, $C = -1$

and hence the particular integral is $y_p = -\frac{1}{2}e^x + 2\sin x - \cos x$

Thus the general solution of the given differential is

$$y = y_c + y_p = c_1e^{3x} + c_2e^{-x} - \frac{1}{2}e^x + 2\sin x - \cos x \quad \text{(Ans)}$$

Example-2 Solve the differential equation by the method of undetermined coefficients:

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x^2e^x$$

Solution Given differential equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x^2e^x$ (1)

The corresponding homogeneous equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$ (2)

Let $y = e^{mx}$ be a trial solution of the equation (2), then the auxiliary equation is

$$m^2 - 3m + 2 = 0 \Rightarrow (m-2)(m-1) = 0 \therefore m = 1, 2$$

Thus the complementary function of (1) is $y_c = c_1e^x + c_2e^{2x}$ (3)

The non-homogeneous term $F(x) = x^2e^x$ is the one UC function given by x^2e^x .

(a) Form the UC set for one UC function x^2e^x . We have $S = \{x^2e^x, xe^x, e^x\}$

(b) By examining the complementary function, we observe that $S = \{x^2e^x, xe^x, e^x\}$ includes e^x , which is included in the complementary function and so is a solution of the corresponding homogeneous differential equation. Thus we multiply each member of S by x to obtain the revised UC set $S' = \{x^3e^x, x^2e^x, xe^x\}$.

Let the particular solution of (1) be $y_p = Ax^3e^x + Bx^2e^x + Cxe^x$ (4)

$$\begin{aligned} \text{Then, } y_p' &= 3Ax^2e^x + Ax^3e^x + 2Bxe^x + Bx^2e^x + Ce^x + Cxe^x = Ax^3e^x + (3A+B)x^2e^x + (2B+C)xe^x + Ce^x \\ y_p'' &= Ax^3e^x + 3Ax^2e^x + (3A+B)x^2e^x + (6A+2B)xe^x + (2B+C)xe^x + (2B+C)e^x + Ce^x \\ &= Ax^3e^x + (6A+B)x^2e^x + (6A+4B+C)xe^x + (2B+2C)e^x \end{aligned}$$

Substituting these in (1) we obtain,

$$\begin{aligned} &\{Ax^3e^x + (6A+B)x^2e^x + (6A+4B+C)xe^x + (2B+2C)e^x\} - 3\{Ax^3e^x + (3A+B)x^2e^x \\ &\quad + (2B+C)xe^x + Ce^x\} + 2(Ax^3e^x + Bx^2e^x + Cxe^x) = x^2e^x \\ \Rightarrow &-3Ax^2e^x + (6A-2B)xe^x + (2B-C)e^x = x^2e^x \end{aligned}$$

The above relation is an identity for all x and hence equating the coefficients of like terms on both sides, we get $-3A = 1$, $6A - 2B = 0$, $2B - C = 0$

Solving above equations, we find that $A = -\frac{1}{3}$, $B = -1$, $C = -2$

and hence the particular integral is $y_p = -\frac{1}{3}x^3e^x - x^2e^x - 2xe^x$

Thus the general solution of the given differential is $y = y_c + y_p = c_1e^x + c_2e^{2x} - \frac{1}{3}x^3e^x - x^2e^x - 2xe^x$ (**Ans**)

Example-3 Solve by the method of undetermined coefficient $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = e^x$

(Ans) $y = (c_1 + c_2x)e^{3x} + \frac{1}{4}e^x$

Example-4 Solve by the method of undetermined coefficients $(D^2 + 2D + 2)y = 2e^{-x}$

(Ans.) $y = e^{-x}(c_1 \cos x + c_2 \sin x) + 2e^{-x}$

Example-5 Solve by the method of undetermined coefficient $(D^2 - 1)y = e^{2x}$

(Ans.) $y = c_1 e^{-x} + c_2 e^x + \frac{1}{3} e^{2x}$

Example-6 Solve by the method of undetermined coefficient $(D^2 + 16)y = 48 \cos x$

(Ans.) $y = c_1 \cos 4x + c_2 \sin 4x + \frac{16}{5} \cos x$

Example-7 Solve the following differential equation by the method of undetermined coefficient

$$(D^2 - 2D + 1)y = x \sin x$$

Solution Given equation $(D^2 - 2D + 1)y = x \sin x$... (1)

Let us consider the homogeneous equation

$$(D^2 - 2D + 1)y = 0 \quad \dots (2)$$

Let $y = e^{mx}$ be a trial solution of the equation (2), then the auxiliary equation is

$$m^2 - 2m + 1 = 0 \quad \Rightarrow (m - 1)^2 = 0 \quad \therefore m = 1, 1$$

$\therefore$ The complementary solution of (1) is $y_c = (c_1 + c_2 x)e^x$,

where c_1, c_2 are arbitrary constants.

The non-homogeneous terms $F(x) = x \sin x$ is the one UC function given by $x \sin x$

Form the UC set for one UC function is given by $x \sin x$, we have

$$S = \{x \sin x, x \cos x, \sin x, \cos x\}$$

Let the Particular solution of (1) be

$$y_p = Ax \sin x + Bx \cos x + C \sin x + E \cos x \quad \dots (3)$$

$$\therefore y'_p = A \sin x + Ax \cos x + B \cos x - Bx \sin x + C \cos x - E \sin x$$

$$\text{and } y''_p = A \cos x + A \cos x - Ax \sin x - B \sin x - B \sin x - Bx \cos x - C \sin x - E \cos x$$

$$= 2A \cos x - Ax \sin x - 2B \sin x - Bx \cos x - C \sin x - E \cos x$$

Substituting these results in (1), we obtain

$$2A \cos x - Ax \sin x - 2B \sin x - Bx \cos x - C \sin x - E \cos x - 2A \sin x$$

$$- 2Ax \cos x - 2B \cos x + 2Bx \sin x - 2C \cos x + 2E \sin x + Ax \sin x + Bx \cos x + C \sin x + E \cos x = x \sin x$$

$$\Rightarrow (2A - 2B - 2C) \cos x + (-2B - 2A + 2E) \sin x + 2Bx \sin x - 2Ax \cos x = x \sin x$$

Equating coefficients of like terms from both sides, we get

$$A = 0,$$

$$2B = 1 \Rightarrow B = \frac{1}{2}$$

$$2A - 2B - 2C = 0$$

$$\Rightarrow 0 - 1 - 2C = 0 \Rightarrow C = -\frac{1}{2}$$

$$\text{and } -2B - 2A + 2E = 0$$

$$\Rightarrow -1 - 0 + 2E = 0 \Rightarrow E = \frac{1}{2}$$

Now putting the values of A, B, C, E in (3), we get

$$y_p = \frac{1}{2}x \cos x - \frac{1}{2} \sin x + \frac{1}{2} \cos x$$

$$\Rightarrow y_p = \frac{1}{2}(x \cos x + \cos x - \sin x)$$

$\therefore$ The general solution is

$$y = y_c + y_p$$

$$\Rightarrow y = (c_1 + c_2 x)e^x + \frac{1}{2}(x \cos x + \cos x - \sin x) \quad \text{(Ans.)}$$

Example-8 Solve by the method of undetermined coefficient

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = e^x \sin x$$

$$\text{(Ans.) } y = c_1 + c_2 e^{2x} - \frac{1}{2} e^x \sin x$$

Example-9 Solve by the method of undetermined coefficient

$$(D^2 - 3D)y = 8e^{3x} + 4 \sin x$$

$$\text{(Ans.) } y = c_1 + c_2 e^{3x} + \frac{8}{3} x e^{3x} - \frac{2}{5} \sin x + \frac{6}{5} \cos x$$

Example-10 Solve by the method of undetermined coefficient

$$(D^3 + 2D - D - 2)y = e^x + x^2$$

$$\text{(Ans.) } y = (c_1 + c_2 x)e^{2x} + \frac{1}{20} x^5 e^{2x} + \frac{1}{6} x^3 e^{2x}$$

Example-12 Solve the equation by the method of undetermined coefficients:

$$\frac{d^3 y}{dx^3} - 3\frac{d^2 y}{dx^2} + 4\frac{dy}{dx} - 2y = e^x + \cos x$$

Solution Given equation may be written as

$$(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x \quad \dots(1)$$

The corresponding homogeneous equation is

$$(D^3 - 3D^2 + 4D - 2)y = 0 \quad \dots(2)$$

Let $y = e^{mx}$ be a trial solution of (2), then the auxiliary equation is

$$m^3 - 3m^2 + 4m - 2 = 0 \quad [\because e^{mx} \neq 0]$$

$$\Rightarrow m^3 - m^2 - 2m^2 + 2m + 2m - 2 = 0$$

$$\Rightarrow m^2(m-1) - 2m(m-1) + 2(m-1) = 0$$

$$\Rightarrow (m-1)(m^2 - 2m + 2) = 0$$

$$\therefore m-1=0 \text{ or, } (m^2 - 2m + 2) = 0$$

$$\Rightarrow m=1 \text{ or, } m = \frac{2 \pm \sqrt{4-8}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$$

$$\therefore y_c = c_1 e^x + e^x(c_2 \cos x + c_3 \sin x)$$

Here undetermined coefficient function is $F(x) = e^x + \cos x$

Undetermined coefficient set of e^x is $S_1 = \{e^x\}$

Undetermined coefficient set of $\cos x$ is $S_2 = \{\cos x, \sin x\}$

Since the element e^x of S_1 is also in y_c , so multiplying the elements of S_1 by x , we get

$$S'_1 = \{xe^x\} \text{ and } S_2 = \{\cos x, \sin x\}$$

$\therefore$ The final UC set $S = \{xe^x, \cos x, \sin x\}$

$\therefore$ Particular solution of (1) is

$$y_p = Axe^x + B\cos x + C\sin x \quad \dots (3)$$

$$\therefore y'_p = A(xe^x + e^x) - B\sin x + C\cos x$$

$$y''_p = A(xe^x + 2e^x) - B\cos x - C\sin x$$

$$\text{and } y'''_p = A(xe^x + 3e^x) + B\sin x - C\cos x$$

Substituting these results in (1), we obtain

$$A(xe^x + 3e^x) + B\sin x - C\cos x - 3A(xe^x + 2e^x) + 3B\cos x + 3C\sin x$$

$$+4A(xe^x + e^x) - 4B\sin x + 4C\cos x - 2Axe^x - 2B\cos x$$

$$-2C\sin x = e^x + \cos x$$

$$\Rightarrow Ae^x + (B + 3C - 4B - 2C)\sin x + (-C + 3B + 4C - 2B)\cos x = e^x + \cos x$$

$$\Rightarrow Ae^x + (C - 3B)\sin x + (B + 3C)\cos x = e^x + \cos x$$

Equating like terms from both sides we get.

$$A = 1,$$

$$C - 3B = 0 \Rightarrow C = 3B$$

$$\text{and } B + 3C = 1 \Rightarrow B + 3(3B) = 1 \Rightarrow 10B = 1$$

$$\therefore B = \frac{1}{10} \text{ and } C = \frac{3}{10}$$

Putting the values of A , B and C in (3), we obtain the particular integral

$$y_p = xe^x + \frac{1}{10}\cos x + \frac{3}{10}\sin x$$

$\therefore$ The general solution is

$$y = y_c + y_p$$

$$\therefore y = ce^x + e^x(c_2 \cos x + c_3 \sin x) + xe^x + \frac{1}{10}\cos x + \frac{3}{10}\sin x \text{ (Ans.)}$$

Section-7.2

Method of Undetermined Coefficients (Annihilator Approach)

7.2.1 Method of Undetermined Coefficients (Annihilator Approach)

Let us consider a differential equation

$$f(D)y = F(x) \quad \dots (1)$$

where the operator $f(D)$ is a polynomial in the operator D is given by

$$f(D) = a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n \quad \dots (2)$$

Let the roots of the auxiliary equation $f(m) = 0$ of the homogeneous equation $f(D)y = 0$ be

$$m = m_1, m_2, \dots, m_n \quad \dots (3)$$

The general solution of (1) is $y = y_c + y_p$... (4)

where y_c can be obtained at once from the values of m in (3) and y_p is any particular solution of (1), which yet to be obtained.

Now suppose that the right member $F(x)$ of (1) is itself a particular solution of some homogeneous linear differential equation with constant coefficients, $L(D)F(x) = 0$... (5)

whose auxiliary equation has the roots

$$m' = m'_1, m'_2, \dots, m'_n \quad \dots (6)$$

Recall that the values of m' in (6) or $L(D)$ in (5) can be obtained by inspection from $F(x)$.

Now applying differential operator $L(D)$ to both sides of (1), we obtain the differential equation

$$L(D)f(D)y = 0 \quad \dots (7)$$

which has as the roots of its auxiliary equation the values of m from (3) and m' from (6).

Hence the general solution of (7) contains the y_c of (4) and so is of the form $y = y_c + y_q$

But also any particular solution of (1) must satisfy (7). Now, if

$$\begin{aligned} f(D)(y_c + y_q) &= F(x) \\ \Rightarrow f(D)y_q &= F(x) \quad [\because f(D)y_c = 0] \end{aligned}$$

Then deleting the y_c from the general solution of (7) leaves a function y_q that for some numerical values of its coefficients must satisfy (1); that is, the coefficients in y_q can be determined so that $y_q = y_p$. These numerical coefficients can be determined by equating the coefficients of like terms from both sides of the equation $f(D)y_q = F(x)$.

Note: This method is applicable when, and only when, the right member of the equation is itself a particular solution of some homogeneous linear differential equation with constant coefficients.

7.2.2 Annihilator Operators

Let us consider a general higher order differential equation

$$f(D)y = F(x) \quad \dots (1)$$

where the operator $f(D)$ is given by $f(D) = a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n$... (2)

If the operator $L(D)$ vanishes $F(x)$, i.e., $L(D)F(x) = 0$, then the operator $L(D)$ is called annihilator operator of $F(x)$.

Examples:

(i) Let us consider a differential equation $f(D)y = x^2$.

Since the operator D^3 vanishes the function $F(x) = x^2$, i.e., $D^3\{F(x)\} = D^3(x^2) = 0$, then the operator $L(D) = D^3$ is the annihilator operator of $F(x) = x^2$.

(ii) Let us consider a differential equation $f(D)y = e^{2x}$.

Since the operator $(D-2)$ vanishes the function $F(x) = e^{2x}$, i.e., $(D-2)\{F(x)\} = (D-2)(e^{2x}) = 0$, then the operator $L(D) = (D-2)$ is the annihilator operator of $F(x) = e^{2x}$.

(iii) Let us consider a differential equation $f(D)y = \sin 2x$.

Since the operator $(D^2 + 4)$ vanishes the function $F(x) = \sin 2x$, i.e., $(D^2 + 4)\{F(x)\} = (D^2 + 4)(\sin 2x) = 0$, then the operator $L(D) = (D^2 + 4)$ is the annihilator operator of $F(x) = \sin 2x$.

7.2.3 List of Some Annihilator Operators

A constant function $y = c$ is annihilated by D , since $D(c) = 0$. The function $y = x$ is annihilated by the differential operator D^2 , since $Dx = 1$ and $D^2x = 0$. Similarly, $D^3x^2 = 0$, and so on.

Some operators, which annihilate the specific functions, are given in the table below:

Operators	Functions	Annihilation Operations
D^n	x^{n-1}	$D^n x^{n-1} = 0$
$(D-a)$	e^{ax}	$(D-a)e^{ax} = 0$
$(D-a)^n$	$x^{n-1}e^{ax}$	$(D-a)^n x^{n-1}e^{ax} = 0$
$(D^2 + a^2)$	$\sin ax$ or, $\cos ax$	$(D^2 + a^2)\sin ax = 0$ or, $(D^2 + a^2)\cos ax = 0$
$(D^2 + a^2)^n$	$x^{n-1}\sin ax$ or, $x^{n-1}\cos ax$	$(D^2 + a^2)^n x^{n-1}\sin ax = 0$ or, $(D^2 + a^2)^n x^{n-1}\cos ax = 0$

If a differential operator annihilate a function of its highest power, then the operator also annihilates all the similar functions of its lower powers.

(i) The differential operator D^n annihilates each of the functions $1, x, x^2, x^3, \dots, x^{n-1}$

(ii) The differential operator $(D-a)^n$ annihilates each of the functions $e^{ax}, xe^{ax}, x^2e^{ax}, x^3e^{ax}, \dots, x^{n-1}e^{ax}$

(iii) The differential operator $\left[D^2 - 2\alpha D + (\alpha^2 + \beta^2)\right]^n = \left[(D-\alpha)^2 + \beta^2\right]^n$ annihilates each of the functions

$$\left\{ \begin{array}{l} e^{\alpha x} \cos \beta x, xe^{\alpha x} \cos \beta x, x^2e^{\alpha x} \cos \beta x, \dots, x^{n-1}e^{\alpha x} \cos \beta x \\ e^{\alpha x} \sin \beta x, xe^{\alpha x} \sin \beta x, x^2e^{\alpha x} \sin \beta x, \dots, x^{n-1}e^{\alpha x} \sin \beta x \end{array} \right\}$$

Solved Problems:

Example-1 Find a differential operator that annihilates the given function.

(i) $4x^4 - 6x^3 + 2x^2 - 5$, (ii) e^{-4x} , (iii) $3e^{3x} + 5xe^{3x} + 2x^2e^{3x}$,

(iv) $4e^{-2x}\sin 3x - 5e^{-2x}\cos 3x$, (v) $5 - x + xe^{-x} + 2\sin x$.

Solution

(i) Since, $D^5x^4 = 0$, the operator D^5 annihilate $4x^4 - 6x^3 + 2x^2 - 5$, i.e., $D^5(4x^4 - 6x^3 + 2x^2 - 5) = 0$.

(ii) Since, $(D+4)e^{-4x} = 0$, the operator $(D+4)$ annihilate e^{-4x} , i.e., $(D+4)e^{-4x} = 0$.

(iii) Since, $(D-3)e^{3x} = 0$, $(D-3)^2xe^{3x} = 0$, $(D-3)^3x^2e^{3x} = 0$, the operator $(D-3)^3$ annihilate linear combination $3e^{3x} + 5xe^{3x} + 2x^2e^{3x}$, i.e., $(D-3)^3(3e^{3x} + 5xe^{3x} + 2x^2e^{3x}) = 0$.

(iv) Since, $(D^2+9)\sin 3x = 0$ or $(D^2+9)\cos 3x = 0$, the operator $\{(D+2)^2+9\} = (D^2+4D+13)$ annihilate linear combination $4e^{-2x}\sin 3x - 5e^{-2x}\cos 3x$, i.e., $(D^2+4D+13)(4e^{-2x}\sin 3x - 5e^{-2x}\cos 3x) = 0$.

(v) Since, $D^2(5-x) = 0$, $(D+1)^2xe^{-x} = 0$ and $(D^2+1)\sin x = 0$, the operator $D^2(D+1)^2(D^2+1)$ annihilate linear combination $5 - x + xe^{-x} + 2\sin x$, i.e., $D^2(D+1)^2(D^2+1)(5 - x + xe^{-x} + 2\sin x) = 0$.

Example-2 Find a homogeneous linear equation, with constant coefficients, that has as a particular solution $y = -2x + 5e^{4x} - \cos x$.

Solution The term $-2x$ is associated with a double root $m=0, 0$, the term $5e^{4x}$ with $m=4$ and the term $(-\cos x)$ with the pair of imaginary roots $m=0 \pm i$.

Hence the auxiliary equation is $m^2(m-4)(m+i)(m-i) = 0 \Rightarrow (m^3-4m^2)(m^2+1) = 0 \therefore (m^5-4m^4+m^3-4m^2) = 0$

Therefore, the given function is a solution of the differential equation

$$(D^5 - 4D^4 + D^3 - 4D^2)y = 0 \text{ (Ans.)}$$

Example-3 Find a homogeneous linear equation with real, constant

coefficients that is satisfied by $y = 2xe^{-x}\cos 3x$.

Solution The desired equation must have its auxiliary equation with roots

$m = -1 \pm 3i, -1 \pm 3i$. The roots $m = -1 \pm 3i$ correspond to factors $(m+1)^2 + 3^2$,

so the auxiliary equation must be

$$\{(m+1)^2 + 3^2\}^2 = 0$$

$$\Rightarrow (m^2 + 2m + 10)^2 = 0$$

$$\Rightarrow m^4 + 4m^2 + 100 + 4m^3 + 40m + 20m^2 = 0$$

$$\therefore m^4 + 4m^3 + 24m^2 + 40m + 100 = 0$$

Therefore, the given function is a solution of the differential equation

$$(D^4 + 4D^3 + 24D^2 + 40D + 100)y = 0 \text{ (Ans.)}$$

Example-4 Solve the following equation by the method of undetermined coefficients with the help of annihilation operators: $(D^2 - 3D)y = 8e^{3x} + 4\sin x$.

Solution Given equation $(D^2 - 3D)y = 8e^{3x} + 4\sin x$... (1)

The corresponding homogeneous equation is $(D^2 - 3D)y = 0$... (2)

Let $y = e^{mx}$ be a trial solution of (2), then the auxiliary equation is

$$m^2 - 3m = 0 \Rightarrow m(m - 3) = 0 \therefore m = 0, 3.$$

$\therefore$ The complementary solution is $y_c = c_1 + c_2e^{3x}$... (3)

Here undetermined coefficient function is $F(x) = 8e^{3x} + 4\sin x$.

Since the operator $(D - 3)$ annihilates e^{3x} and $(D^2 + 1)$ annihilates $\sin x$,

we apply differential operator $(D - 3)(D^2 + 1)$ to both sides of (1), we get

$$(D - 3)(D^2 + 1)(D^2 - 3D)y = 0 \quad \dots (4)$$

$$[\because (D - 3)8e^{3x} = 0 \text{ \& } (D^2 + 1)4\sin x = 0]$$

The auxiliary equation of (4) is

$$(m - 3)(m^2 + 1)(m^2 - 3m) = 0$$

$$\Rightarrow m(m - 3)^2(m^2 + 1) = 0$$

$$\therefore m = 0, 3, 3, \pm i$$

Thus the general solution of (4) is

$$y = c_1 + (c_2 + c_3x)e^{3x} + c_4 \cos x + c_5 \sin x$$

Since the terms c_1 and c_2e^{3x} also include in (3), excluding these terms we form the particular solution of the form

$$y_p = Axe^{3x} + B\cos x + C\sin x \quad \dots (5)$$

where, A , B and C are undetermined coefficients.

Differentiating (5) successively two times, we obtain

$$y'_p = Ae^{3x} + 3Axe^{3x} - B\sin x + C\cos x$$

$$\text{and } y''_p = 3Ae^{3x} + 3Ae^{3x} + 9Axe^{3x} - B\cos x - C\sin x$$

Substituting these results in (1), we get

$$6Ae^{3x} + 9Axe^{3x} - B\cos x - C\sin x - 3(Ae^{3x} + 3Axe^{3x} - B\sin x + C\cos x) = 8e^{3x} + 4\sin x$$

$$\Rightarrow 3Ae^{3x} - (B + 3C)\cos x + (3B - C)\sin x = 8e^{3x} + 4\sin x$$

Equating the coefficients of like terms from both sides, we get

$$3A = 8 \Rightarrow A = \frac{8}{3};$$

$$-B-3C=0 \Rightarrow B=-3C$$

$$3B-C=4 \Rightarrow -9C-C=4 \Rightarrow -10C=4 \Rightarrow C=-\frac{2}{5};$$

$$\text{so, } B=\frac{6}{5}$$

Now putting the values of A, B, C in (5), we get

$$y_p = \frac{8}{3}xe^{3x} + \frac{6}{5}\cos x - \frac{2}{5}\sin x$$

$\therefore$ The general solution is

$$y = y_c + y_p$$

$$\therefore y = c_1 + c_2e^{3x} + \frac{8}{3}xe^{3x} + \frac{6}{5}\cos x - \frac{2}{5}\sin x \quad \textbf{(Ans.)}$$

Example-5 Solve the following equation by the method of undetermined coefficients with the help of annihilation operators:

$$(D^2 - 2D - 3)y = 2e^x + 20\sin x$$

Solution Given equation is $(D^2 - 2D - 3)y = 2e^x + 20\sin x \quad \dots (1)$

The corresponding homogeneous equation is $(D^2 - 2D - 3)y = 0 \quad \dots (2)$

Let $y = e^{mx}$ be a trial solution of (2), then the auxiliary equation is

$$m^2 - 2m - 3 = 0 \quad [\because e^{mx} \neq 0]$$

$$\Rightarrow m^2 - 3m + m - 3 = 0$$

$$\Rightarrow m(m-3) + 1(m-3) = 0$$

$$\Rightarrow (m+1)(m-3) = 0$$

$$\Rightarrow m = -1, 3$$

$$\therefore y_c = c_1e^{-x} + c_2e^{3x} \quad \dots (3)$$

Here undetermined coefficient function is $F(x) = 2e^x + 20\sin x$ is a particular solution of a homogeneous linear differential equation whose auxiliary equation has the roots

$$m' = 1, \pm i$$

Therefore, the function F is a particular solution of the equation

$$(D-1)(D^2+1)F = 0$$

Now we apply differential operator $(D-1)(D^2+1)$ to both sides of (1), we get

$$(D-1)(D^2+1)(D^2-2D-3)y = 0 \quad \dots (4)$$

$$[\because (D-1)2e^x = 0 \text{ \& } (D^2+1)20\sin x = 0]$$

The auxiliary equation of (4) is

$$(m-1)(m^2+1)(m^2-2m-3)=0$$

$$\Rightarrow (m-1)(m^2+1)(m+1)(m-3)=0$$

$$\therefore m = -1, 1, 3, \pm i$$

Thus the general solution of (4) is $y = c_1 e^{-x} + c_2 e^{3x} + c_3 e^x + c_4 \sin x + c_5 \cos x$

Since the terms $c_1 e^{-x}$ and $c_2 e^{3x}$ include in (3) also, excluding these terms we form the particular solution of the form

$$y_p = Ae^x + B \sin x + C \cos x \quad \dots (5)$$

where, A, B and C are undetermined coefficients.

Differentiating (5) successively two times, we obtain

$$y'_p = Ae^x + B \cos x - C \sin x$$

$$\text{and } y''_p = Ae^x - B \sin x - C \cos x$$

Substituting these results in (1), we get

$$Ae^x - B \sin x - C \cos x - 2(Ae^x + B \cos x - C \sin x) - 3(Ae^x + B \sin x + C \cos x) = 2e^x + 20 \sin x$$

$$\Rightarrow (A - 2A - 3A)e^x + (-B + 2C - 3B) \sin x + (-C - 2B - 3C) \cos x = 2e^x + 20 \sin x$$

$$\Rightarrow -4Ae^x + (2C - 4B) \sin x - (2B + 4C) \cos x = 2e^x + 20 \sin x$$

Equating like terms from both sides, we get

$$-4A = 2 \Rightarrow A = -1/2$$

$$-4B + 2C = 20$$

$$\Rightarrow -2B + C = 10 \quad \dots (6)$$

$$\text{and } 2B + 4C = 0 \quad \dots (7)$$

$$\therefore (4) + (5) \Rightarrow 5C = 10 \Rightarrow C = 2$$

$$\text{then } 2B = -4C \Rightarrow B = -4$$

Now putting the values of A, B, C in (3), we get $y_p = \frac{1}{2}e^x - 4 \sin x + 2 \cos x$

$\therefore$ The general solution is $y = y_c + y_p$

$$\text{or, } y = c_1 e^{-x} + c_2 e^{3x} - \frac{1}{2}e^x - 4 \sin x + 2 \cos x \quad \textbf{(Ans.)}$$